

BST Physics Curriculum Vol 11 Chapter 6 — K3 Surfaces + Niemeier Lattices + Leech v0.4 SIGNATURE (refilled per Cal #104)

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Vol 11 Chapter 6 — K3 Surfaces + Niemeier Lattices + Leech

Chapter motivation

Algebraic geometry + lattice theory + sporadic finite groups intersect in remarkable ways: - K3 surface (Kodaira-Spencer 1956 dimension-2 Calabi-Yau): smooth simply-connected complex surface with trivial canonical bundle + $h^{\{1,0\}} = 0$ + $h^{\{2,0\}} = 1$; Hodge diamond $1, 0, 1, 0, 20, 0, 1, 0, 1$ with $h^{\{1,1\}} = 20$ - Niemeier lattices (Niemeier 1973): 24 even unimodular Euclidean lattices in dim 24, classified by root systems - Leech lattice (Conway 1968): unique Niemeier lattice without roots; automorphism group $\text{Co}_0 = 2 \cdot \text{Co}_1$ (sporadic simple) - Mathieu group M_{24} (Mathieu 1861-1873): sporadic finite simple group order 244823040; $\subset \text{Co}_0$

BST connection: per Vol 0 Ch 9 Strong-Uniqueness C11 (RIGOROUSLY CLOSED via T2440 Thursday Session 3): D_{IV}^5 uniquely supports 5-family Bridge Object architecture with 16 effective members across 5 families — K3 family (Family 4) + Heegner-trio (Family 1) + $\chi=24$ non-Heegner (Family 2) + N_{max} -anchored (Family 3) + Q^5 -family (Family 5). K3 surface + Mathieu M_{24} + Leech connections appear in Family 4 (K3 + K77 PATH B RATIFIED status) + Family 2 ($\chi=24$ substrate-anchor cluster).

Section 6.0b — Reader-grade 3-level pedagogy

Level 1: K3 surfaces (algebraic geometry; Hodge diamond $h^{\{1,1\}} = 20$) + Niemeier lattices (24 even unimodular dim 24) + Leech lattice (Conway 1968; unique no-root Niemeier) + Mathieu M_{24} sporadic group; BST cross-link: 5-family Bridge Object architecture per Vol 0 Ch 9 C11 RIGOROUSLY CLOSED + T2440 (D_{IV}^5 uniquely supports).

Level 2 (graduate-mathematician): K3 surface (Kodaira 1956 dimension-2 Calabi-Yau): smooth simply-connected complex surface with trivial canonical bundle $K_X \cong \mathbb{Q}_X$; $h^1(X) = 0$, $h^2(X) = 1$. Hodge diamond: $h^{0,0} = h^{2,2} = 1$, $h^{1,0} = h^{0,1} = h^{2,1} = h^{1,2} = 0$, $h^{2,0} = h^{0,2} = 1$, $h^{1,1} = 20$; total Hodge number $24 = 2 + 20 + 2$. Niemeier lattices (Niemeier 1973): 24 even unimodular Euclidean lattices in dim 24 classified by their root systems (A_1^{24} , A_2^{12} , A_3^8 , ..., E_8^3 , $E_7^2 D_{10}$, etc.); 23 with non-trivial roots + 1 unique no-root = Leech lattice Λ_{24} . Conway 1968 discovered Leech lattice as unique Niemeier without roots; Conway groups $Co_0 = \text{Aut}(\Lambda_{24})$ sporadic finite group order 8315553613086720000; $Co_1 = Co_0 / \{\pm 1\}$ sporadic simple. Mathieu M_{24} (Mathieu 1861-1873): sporadic finite simple group order 244823040 = $2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$; $\subset Co_0$ via Leech lattice subgroup; quintuply-transitive on 24 points; closely connected to extended binary Golay code [24, 12, 8]. BST 5-family Bridge Object architecture (Vol 0 Ch 9 C11 RIGOROUSLY CLOSED via T2440 Thursday Session 3): D_{IV}^5 uniquely supports 5 Bridge Object families with 16 effective members. Family 1 Heegner-trio: K47 + K70 + K62 (Cremona elliptic curves at BST primary discriminants). Family 2 $\chi=24$ non-Heegner: K76 Leech + K81 24-cell + K82 $\Delta(\tau)$ modular; $\chi=24 = K3$ Euler characteristic = Mathieu $M_{24} + (Z_2)^4$ group order = $n_C! \cdot 2^{n_C-1}$ (Hua volume formula for D_{IV}^5). Family 3 $N_{\max}$ -anchored: K80 + K84 ($N_{\max}=137$ anchor). Family 4 K3-family: K45 RATIFIED + K77 PATH B RATIFIED-status + K3F5 (K3 surface Hodge + 5-family extension). Family 5 Q^5 -family: 6 geometric-enumerated members per Grace Mode 6 enumeration (Q^5 Chern integers all BST primary: $c_1=n_C$, $c_2=11$, $c_3=13$, $c_4=9$, $c_5=C_2=6$). 3 K57 RATIFIED central hubs ($K3 + 49a1 + Q^5$) + 16 family-members + 3 RATIFIED-status sub-members = 19 effective Bridge Object structural positions. T2440 RIGOROUSLY CLOSED: only D_{IV}^5 at $\dim_C = 5$ supports this 5-family architecture (alt-HSDs fail).

Level 3 (5th-grader accessible): K3 surfaces (algebraic geometry) + Niemeier lattices (24 even integer lattices in 24D) + Leech lattice (Conway 1968; unique no-root Niemeier) + Mathieu M_{24} (sporadic finite group) all connect to BST via the 5-family Bridge Object architecture (Vol 0 Ch 9 C11 RIGOROUSLY CLOSED). The Hua volume formula for D_{IV}^5 uses $|\Gamma| = n_C! \cdot 2^{n_C-1} = 120 \cdot 16 = 1920$ — same Weyl group structure as Mathieu $M_{24} + K3$ Euler characteristic $\chi = 24$. The 5-family Bridge Object architecture (16 effective members across 5 families) uniquely supports D_{IV}^5 via T2440.

Section 6.1 — K3 Surface (Kodaira 1956)

Smooth simply-connected complex surface; trivial canonical bundle $K_X \cong \mathbb{Q}_X$; $h^1(X) = 0$; $h^2(X) = 1$.

Hodge diamond: 1, 0, 1, 0, 20, 0, 1, 0, 1 with $h^{1,1} = 20$. Total Hodge = 24 = Euler characteristic $\chi(K3)$.

Section 6.2 — Niemeier Lattices (1973)

24 even unimodular Euclidean lattices in dim 24, classified by root systems (A_1^{24} , A_2^{12} , ..., E_8^3 , $E_7^2 D_{10}$, etc.).

23 with non-trivial roots + 1 unique no-root = Leech lattice Λ_{24} .

Section 6.3 — Leech Lattice + Conway Groups (1968)

Conway 1968: Leech lattice unique Niemeier without roots; Conway groups $Co_0 = \text{Aut}(\Lambda_{24})$ (order 8315553613086720000) + $Co_1 = Co_0 / \{\pm 1\}$ (sporadic simple).

Section 6.4 — Mathieu M_{24} (1861-1873)

Mathieu M_{24} sporadic finite simple group order $244823040 = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$; $\subset Co_0$; quintuply-transitive on 24 points; connected to extended binary Golay code [24, 12, 8].

Section 6.5 — BST 5-Family Bridge Object Architecture (T2440 RIGOROUSLY CLOSED)

Vol 0 Ch 9 Strong-Uniqueness C11 RIGOROUSLY CLOSED via T2440 (Lyra Thursday Session 3): D_{IV}^5 uniquely supports 5-family Bridge Object architecture with 16 effective members across 5 families.

Family 1 Heegner-trio: $K47 + K70 + K62$ Cremona elliptic curves at BST primary discriminants. Family 2 $\chi=24$ non-Heegner: $K76$ Leech + $K81$ 24-cell + $K82 \Delta(\tau)$; $\chi=24 = K3 \chi = M_{24} \cdot (Z_2)^4$ + Hua volume formula $|\Gamma| = n_C! \cdot 2^{\{n_C-1\}} = 1920$. Family 3 $N_{\max}$ -anchored: $K80 + K84$. Family 4 K3-family: $K45$ RATIFIED + $K77$ PATH B RATIFIED + $K3F5$. Family 5 Q^5 -family: 6 geometric-enumerated members via Grace Mode 6 (Q^5 Chern integers all BST primary).

3 $K57$ RATIFIED central hubs + 16 family-members + 3 RATIFIED-status sub-members = 19 effective Bridge Object structural positions.

Section 6.6 — Hua Volume Formula = Mathieu/K3 Connection

$|\Gamma| = n_C! \cdot 2^{\{n_C-1\}} = 120 \cdot 16 = 1920$ Weyl group of D_{IV}^5 .

$\text{Vol}(D_{IV}^5) = \pi^{\{n_C\}} / |\Gamma| = \pi^5 / 1920$ per Hua 1958.

Connects to: Mathieu M_{24} + K3 Euler characteristic $\chi = 24$ + baryon circuit orbit (1920 configurations).

Section 6.7 — Honest scope + Connection

- Standard K3 + Niemeier + Leech + Mathieu M_{24} [?]
- BST 5-family Bridge Object architecture (T2440 RIGOROUSLY CLOSED) [?]

- Hua volume formula $|\Gamma| = 1920$ connects D_{IV}^5 to Mathieu/K3
- Open scope: explicit K3-Hodge-on- D_{IV}^5 substrate-derivation (multi-week)

Connection: - Vol 0 Ch 9 Strong-Uniqueness C11 + T2440 - Vol 4 Ch 10 Dark Matter (Weyl group $|\Gamma| = 1920$ cross-link) - Vol 11 Ch 8 Monster Moonshine + Supersingular Primes (Mathieu/Leech extensions)

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